Clustered Safety Events and the Effective Sample Size in Autonomous Vehicle Fleet Rate Estimation

Pouria Emami Tabrizi

September 2026

Abstract

Autonomous vehicle fleet validation estimates rare-event rates — safety-relevant disengagements, near-misses, contact events — per mile driven, and attaches confidence statements to those estimates. The estimators in current use treat logged events as independent arrivals of a Poisson process. Real fleet events are not independent: a construction zone, a rain hour, or a single poorly handled intersection produces several correlated events. We model the event stream as a compound Poisson (Neyman–Scott) process and show that ignoring this clustering inflates the variance of the rate estimator by a design effect $D = E[K^2]/E[K]$, where K is the cluster size. Three consequences follow. First, a nominal 95% confidence interval has asymptotic coverage $2\Phi(z_{0.975}/\sqrt{D}) - 1$, which falls to 74% at $D = 3$ and 54% at $D = 7$. Second, the mileage required to demonstrate that a fleet's event rate lies below a safety target is understated by exactly the factor D, which suggests that standard mileage calculations may understate the evidence required. Third, empirical-Bayes shrinkage weights of the form $n/(n + k)$ over-weight noisy local estimates, costing up to 7.8× in mean squared error in our study. We give a correction that substitutes an effective sample size $n_{\text{eff}} = n/\hat{D}$, where $\hat{D}$ is estimated from the logs themselves, and show by simulation that it restores nominal coverage across the range studied. **All results in this paper are obtained on synthetic data; no real fleet data is used.** We identify the specific public data sources in which $\hat{D}$ could be measured and describe the procedure, but we do not carry out that measurement here. This is an independent preprint and has not been peer reviewed.

1 Introduction

The central statistical difficulty of autonomous vehicle (AV) safety validation is that the events of interest are rare. Kalra and Paddock [5] showed that demonstrating a fatality rate lower than the human benchmark with reasonable confidence would require hundreds of millions of miles, a result that has shaped a decade of subsequent work. Liu and Feng [8] give the modern statement of the problem as a "curse of rarity": the information content per mile is so low that direct estimation is impractical without variance reduction, and much recent effort has gone into importance sampling and scenario-based testing to recover statistical power.

Much of this work rests on a modelling assumption that is seldom made explicit and, as far as we are aware, not yet examined directly: that the safety-relevant events logged by a fleet are *independent*. Terres et al. [13] model the count of behavioral events over m miles as Poisson(θm), which embeds independence in the likelihood. Chen et al. [2] assume i.i.d. features and a Poisson sample size in deriving confidence intervals for importance-sampled rate estimates; the authors note explicitly that "how to properly design the run segments to minimize spatial-temporal dependencies is beyond the scope of the current paper." Zhao et al. [14], in a paper arguing for exactly the statistical rigour we are concerned with here, adopt i.i.d. Bernoulli scenario models while acknowledging that the simplification "may not hold due to factors like software updating of the AV or changing environments."

The independence assumption is analytically convenient and in many settings harmless. In this one, we argue, it is unlikely to hold, in a specific and consequential way. Safety-relevant events arrive in clusters. A construction zone that the perception stack handles poorly produces several disengagements in a few minutes. A rain hour degrades detection across an entire shift. One unprotected left turn geometry generates the same near-miss on every pass. The clusters are not a nuisance detail of the logging pipeline; they are how the underlying failure modes express themselves.

This paper makes the consequence precise. We model the event stream as a compound Poisson process, derive the resulting variance inflation, and quantify what it costs. The contribution is deliberately narrow:

1. We show that the naive Poisson rate estimator remains unbiased but that its variance is inflated by a design effect $D = E[K^2]/E[K]$ (Section 3).

2. We give the exact asymptotic coverage of the naive confidence interval as a function of D, and confirm it by simulation (Proposition 1).

3. We show that the mileage required to demonstrate a safety target scales linearly in D (Proposition 2), which is the safety-critical consequence: a fleet using the standard calculation concludes "demonstrated" before the evidence supports it.

4. We give a correction based on an effective sample size that is estimable from fields fleets already log, and show it restores nominal coverage (Sections 4 and 5).

We are explicit about what this is not. The design effect is textbook survey statistics [6, 3], and cluster-robust variance estimation is standard [7]. No new mathematics is claimed. The contribution is to connect that machinery to a setting where it has not yet been applied, to quantify what its absence costs in the regime AV validation operates in, and to note that the correction requires no new instrumentation.

2 Related Work

Fleet-level rate estimation. The line of work most directly affected is the estimation of event rates from fleet logs. Terres et al. [13] develop maximum likelihood procedures for rate estimation under partial event review, using a Poisson model for the event count. Chen et al. [2] construct confidence intervals for importance-sampled rate estimates, assuming independent features. Zhao et al. [14] propose a probability-of-failure-per-scenario metric and i.i.d. Bernoulli models for scenario-based testing. None of these addresses clustering, serial dependence, or effective sample size directly. Two of the three are explicit that dependence lies outside their scope, and it is that explicitness which makes the present extension straightforward to state.

Traffic conflict extremes. A separate and considerably more dependence-aware literature estimates crash risk from traffic conflicts using extreme value theory, beginning with Songchitruksa and Tarko [12]. Zheng and Sayed [15] develop non-stationary Bayesian hierarchical peak-over-threshold models, and more recent work in the same literature extends conditional peak-over-threshold models with self- and cross-exciting structure and with spatio-temporal hierarchies. This machinery explicitly models dependence between extreme observations.

It does not, however, transfer directly to the fleet setting, and the reason matters for the present paper. Self-exciting and conditional peak-over-threshold models describe a single, densely observed excitation stream at one site: an intersection under continuous video observation, where every conflict is recorded and the excitation kernel is identifiable from inter-event times. Fleet validation aggregates sparse events across many routes, operating conditions, and software releases, where the dependence is *nested* — an event sits within a scene, within a drive, within a release — rather than generated by one excitation process in one time series. Nested dependence of this kind is what design effects and cluster-robust variance estimators are built for, and a point-process excitation model is not the natural description of it.

Dependence corrections elsewhere. Extreme value theory has long handled dependent exceedances by declustering, with the extremal index quantifying the degree of clustering [4]. In survey statistics the design effect [6, 3] plays the same role for clustered sampling designs, and generalized estimating equations [7] provide cluster-robust variance estimates for correlated observations. In financial machine learning, López de Prado [9] identifies overlapping event labels as a source of inflated effective sample size and proposes average-uniqueness weighting and sequential bootstrapping as remedies. The present paper connects these adjacent treatments to the fleet rate estimation setting, where the same difficulty arises but the corresponding corrections have not yet been carried over.

3 Problem Formulation

3.1 A clustered event process

Let a fleet accumulate m miles. We model safety-relevant events as a compound Poisson process of Neyman–Scott type [11]. Let cluster centres — distinct difficult contexts, such as one construction zone encountered once, one rain hour, one problematic intersection geometry — arrive as a Poisson process with rate λ_c per mile, so that the number of clusters is $N_c \sim \text{Poisson}(\lambda_c m)$. Each cluster i contributes K_i logged events, with $K_1, K_2, \ldots$ independent and identically distributed on $\{1, 2, \ldots\}$, independent of N_c. The total event count is

$$N = \sum_{i=1}^{N_c} K_i. \tag{1}$$

The quantity validation cares about is the event rate per mile,

$$\theta = \lambda_c \, E[K], \tag{2}$$

estimated by $\hat{\theta} = N/m$. The classical Poisson model is the special case $K \equiv 1$.

3.2 The design effect

By the standard compound Poisson moment identities,

$$E[N] = \lambda_c m \, E[K], \qquad \text{Var}(N) = \lambda_c m \, E[K^2]. \tag{3}$$

The estimator $\hat{\theta} = N/m$ is therefore unbiased for θ, with

$$\text{Var}(\hat{\theta}) = \frac{\lambda_c E[K^2]}{m} = \frac{\theta}{m} \cdot \frac{E[K^2]}{E[K]} = \frac{\theta D}{m}, \qquad D := \frac{E[K^2]}{E[K]}. \tag{4}$$

We call D the design effect, by analogy with clustered survey sampling [6]. It satisfies $D \geq 1$, with equality if and only if every cluster has size one. Writing $\mu_K = E[K]$, an equivalent form is $D = \mu_K + \mathrm{Var}(K)/\mu_K$: clustering hurts both through the mean cluster size and through its dispersion.

An analyst who assumes independence uses $\mathrm{Var}(\hat{\theta}) = \theta/m$ and therefore understates the standard error by a factor of $\sqrt{D}$. Note that the point estimate is unaffected. The damage is entirely to the uncertainty statement — which, in a safety argument, is the part that carries the claim.

Remark 1. *Equation* (4) *is the perfectly-correlated case of Kish's design effect $D = 1 + (\bar{K} - 1)\rho$, corresponding to $\rho = 1$: every event in a cluster is treated as carrying the information of the same underlying failure mode. A process with partial within-cluster correlation interpolates between $D = 1$ and the value in* (4)*, so the figures reported here should be read as the design effect implied by whatever clustering the analyst's own logs exhibit, not as a universal constant.*

3.3 Coverage of the naive interval

Proposition 1 (Undercoverage). *Let $\hat{\theta} = N/m$ and let the naive interval be $\hat{\theta} \pm z_{1-\alpha/2}\sqrt{\hat{\theta}/m}$. As $m \to \infty$ with θ and D fixed, the asymptotic coverage of this interval is*

$$2\,\Phi\left(\frac{z_{1-\alpha/2}}{\sqrt{D}}\right) - 1, \tag{5}$$

where Φ is the standard normal distribution function.

Proof. By the central limit theorem for compound Poisson sums, $(\hat{\theta} - \theta)\sqrt{m/(\theta D)} \Rightarrow \mathcal{N}(0, 1)$. The naive half-width is $z_{1-\alpha/2}\sqrt{\hat{\theta}/m}$, and $\hat{\theta} \to \theta$ in probability, so the half-width is asymptotically $z_{1-\alpha/2}\sqrt{\theta/m}$, which equals $z_{1-\alpha/2}/\sqrt{D}$ true standard errors. Coverage is therefore the probability that a standard normal variate lies within $z_{1-\alpha/2}/\sqrt{D}$ of zero. □

For a nominal 95% interval this gives 87.1% at $D = 1.67$, 74.2% at $D = 3$, and 54.1% at $D = 7$. At the upper end of that range the interval's actual coverage bears little relation to its nominal level.

3.4 Miles required to demonstrate a target

Validation programmes are usually framed as a demonstration: show, with confidence $1-\alpha$, that the fleet's event rate lies below some target $\theta^* > \theta$. Using the upper confidence limit $\hat{\theta} + z_{1-\alpha/2}\sqrt{\mathrm{Var}(\hat{\theta})}$ and solving for the mileage at which it falls below θ^* gives the following.

Proposition 2 (Required mileage). *Under the clustered process, the mileage required for the upper confidence limit to fall below θ^* is*

$$m_{req} = \frac{z_{1-\alpha/2}^2\,\theta\,D}{(\theta^* - \theta)^2}, \tag{6}$$

which exceeds the mileage computed under the independence assumption by exactly a factor of D.

This is the statement with the most direct practical consequence. For design effects in the range studied here, a programme that sizes its mileage under the independence assumption and stops at that figure will have accumulated between one third and one seventh of the evidence implied by its own confidence statement. We note that the discrepancy runs in the less cautious direction.

4 Proposed Correction

The correction is to estimate D from the logs and use it. Suppose the N logged events can be partitioned into C observed clusters with sizes $K_1, \ldots, K_C$. A natural moment estimator of the design effect is

$$\hat{D} = \frac{\sum_{i=1}^{C} K_i^2}{\sum_{i=1}^{C} K_i}, \tag{7}$$

which is the sample analogue of $E[K^2]/E[K]$. The effective sample size is then

$$n_{\text{eff}} = \frac{N}{\hat{D}}, \tag{8}$$

and the corrected interval is

$$\hat{\theta} \pm z_{1-\alpha/2} \frac{\sqrt{N\hat{D}}}{m} = \hat{\theta} \pm z_{1-\alpha/2} \frac{\sqrt{N}}{m}\sqrt{\hat{D}}. \tag{9}$$

The required-mileage formula of Proposition 2 is corrected by the same factor.

Shrinkage weights. The same substitution repairs a second estimator. Sparse per-segment or per-scenario rates are commonly stabilised by shrinking them toward a fleet-level prior with an empirical-Bayes weight $w = n/(n + k)$, so that a segment with few events leans on the prior and a segment with many events leans on its own data. Under clustering, n overstates the independent information available, w is therefore too large, and the estimator leans on a noisier local estimate than the evidence supports. Substituting $w_{\text{eff}} = n_{\text{eff}}/(n_{\text{eff}} + k)$ restores the intended behaviour. This is the same defect López de Prado [9] identifies for overlapping labels in financial machine learning, arriving here by a different route.

What counts as a cluster. The estimator (7) requires that cluster membership be observable. In fleet logging it usually is, at least approximately, because every logged event already carries identifiers that induce a natural grouping: the vehicle and the calendar date, the drive or run segment, the scene or map location, the weather condition, or the software release. Any of these defines a partition; the appropriate choice is the one matching the failure mode under study. We return to this in Section 6.

5 Simulation Study

Setup. We simulate the process of Section 3 with a true rate of $\theta = 10^{-4}$ events per mile (one per ten thousand miles) over a 5,000,000 mile deployment, with 20,000 replications per configuration. Cluster sizes are drawn from a geometric distribution on $\{1, 2, \ldots\}$ with parameter p, giving mean cluster size $1/p$ and design effect $D = (2 - p)/p$. All data are synthetic; the accompanying script `sim_clustered_av_rates.py` reproduces every figure reported here.

Coverage. Table 1 reports the actual coverage of a nominal 95% interval. The simulated coverage of the naive interval agrees with Proposition 1 to within one percentage point throughout, and the corrected interval holds its nominal level across the whole range. Figure 1 shows the same result over a finer grid.

A mean cluster size of 2 — two logged events per difficult context, which is not an extreme value for fleet data — already reduces a nominal 95% interval to 73% actual coverage.

Table 1: Actual coverage of a nominal 95% confidence interval for the fleet event rate. Synthetic data; 20,000 replications per row.

Mean cluster size	D	Naive (sim.)	Naive (theory)	Corrected	CI width ratio
1.00	1.00	94.8%	95.0%	94.8%	1.00×
1.33	1.67	86.9%	87.1%	94.9%	1.29×
2.00	3.00	73.2%	74.2%	94.5%	1.73×
2.86	4.71	63.0%	63.3%	94.4%	2.17×
4.00	7.00	53.6%	54.1%	94.6%	2.64×

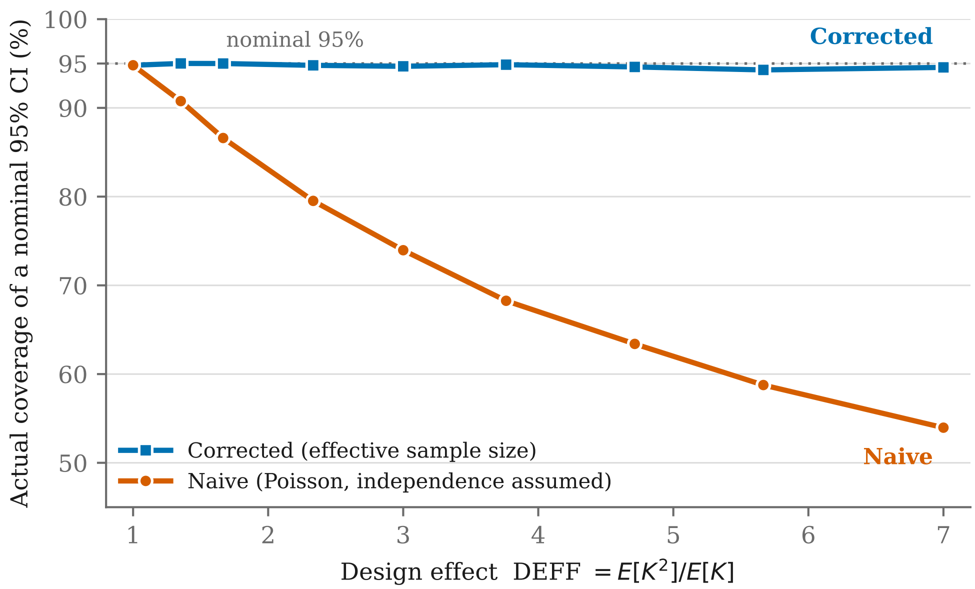

Figure 1: Actual coverage of a nominal 95% confidence interval as a function of the design effect. The naive interval degrades steadily; the effective-sample-size correction holds the nominal level. Synthetic data.

Required mileage. Table 2 reports the mileage needed for the upper confidence limit to fall below a target of 1.5θ. The ratio is exactly D, as Proposition 2 requires.

Table 2: Mileage required to demonstrate that the event rate lies below $\theta^* = 1.5\theta$ at 95% confidence, with $\theta = 10^{-4}$ per mile.

Mean cluster size	D	Assuming independence	Corrected	Shortfall
1.00	1.00	153,658	153,658	1.00×
1.33	1.67	153,658	256,097	1.67×
2.00	3.00	153,658	460,975	3.00×
2.86	4.71	153,658	724,389	4.71×
4.00	7.00	153,658	1,075,608	7.00×

Shrinkage. Table 3 reports the mean squared error of a per-segment rate blended toward an unbiased fleet-level prior with $k = 15$, over 200,000 miles. Because the prior is unbiased by construction, the entire gap is variance: using the nominal event count costs up to 7.8× in mean squared error relative to using the effective count.

Table 3: Mean squared error of an empirical-Bayes blended per-segment rate, using the nominal event count n versus the effective count $n_{\text{eff}} = n/\hat{D}$, with $k = 15$.

Mean cluster size	D	MSE, nominal n	MSE, effective n	Ratio
1.00	1.00	1.647×10^{-10}	1.647×10^{-10}	1.00×
1.33	1.67	2.766×10^{-10}	1.757×10^{-10}	1.57×
2.00	3.00	5.172×10^{-10}	1.785×10^{-10}	2.90×
2.86	4.71	8.658×10^{-10}	1.774×10^{-10}	4.88×
4.00	7.00	1.292×10^{-9}	1.655×10^{-10}	7.81×

6 Estimating the Design Effect in Practice

Nothing in Section 4 requires new instrumentation, and this is the practical argument for the correction: $\hat{D}$ is computable from records that already exist, including public ones.

California DMV disengagement reports. Manufacturers testing autonomous vehicles on California public roads file annual disengagement reports [1]. The published CSV files contain one row per disengagement, with fields for the manufacturer, the date, the vehicle identification number, whether a driver was present, what initiated the disengagement, the road type, and a free-text description of the cause. The date and vehicle identifier together define a vehicle-day, which is a natural and conservative cluster unit: events logged by the same vehicle on the same day are candidates for sharing an underlying context. Computing (7) over vehicle-day clusters, per manufacturer and per year, would yield an empirical distribution of design effects for real fleets.

NHTSA Standing General Order reporting. Incident reports filed under Standing General Order 2021-01 [10] carry date, location, and reporting entity fields that support a coarser but broader clustering by location and reporting period.

We have deliberately not carried out these computations here. Doing so requires care that is beyond the scope of a methodological note: reporting practice varies substantially across manufacturers and years, some filings aggregate rather than enumerate, and the choice of cluster unit is itself a modelling decision that should be justified against the failure mode under study rather than chosen for convenience. We flag this as the single most valuable extension of this work, and we note that it would convert every design effect in Section 5 from an assumed value into a measured one.

7 Discussion and Limitations

What this establishes. That the independence assumption commonly used in AV fleet rate estimation is not without cost; that this cost is a computable function of the clustering in the data; and that the correction is available at no instrumentation cost. The point estimate is unaffected throughout — what changes is the confidence attached to it, and the mileage at which a demonstration can honestly be declared complete.

What this does not establish. It does not establish that any particular fleet's design effect takes any particular value. Every number in Section 5 is generated from an assumed cluster-size distribution. We have not measured D on real data, and until someone does, the practical magnitude of the problem in any specific programme is unknown.

Direction of the error. Under the process modelled here, ignoring clustering makes uncertainty statements overconfident, which is the unsafe direction. We emphasise that this follows from the model, not from a general principle: a process in which logging systematically *suppresses* repeated events within a context, for example through deduplication in the event detection pipeline, could push in the opposite direction. Analysts should determine the sign for their own pipeline rather than assume it.

Cluster identifiability. The correction assumes cluster membership is observable. Where scene or segment boundaries are ambiguous — and they often are, since "the same difficult context" is not a field any logger writes directly — $\hat{D}$ is itself estimated with error, and we do not propagate that uncertainty into the interval. A fully Bayesian treatment placing a prior over the partition would address this and is left to future work. A pragmatic interim measure is sensitivity analysis across several plausible cluster definitions.

Model scope. We assume cluster sizes are independent and identically distributed and independent of the cluster arrival process. Real fleets violate this: difficult contexts are more common in some operational design domains than others, and a software release changes both the arrival rate and the size distribution at once. A hierarchical model with release-level and domain-level effects is the natural extension.

Relationship to existing practice. Fleets are of course aware that their data is structured; engineering teams routinely analyse events by scene, by route, and by release. Our claim is narrower: that this structure does not appear to be propagated into the published confidence statements and mileage targets, and that propagating it changes those statements materially. We would be glad to be shown otherwise for any particular programme.

Acknowledgments

Portions of the literature review and drafting were assisted by Claude (Anthropic); the research direction, modelling choices, and conclusions are the author's own.

References

[1] California Department of Motor Vehicles. Autonomous vehicle disengagement reports, 2024. `https://www.dmv.ca.gov/portal/vehicle-industry-services/autonomous-vehicles/disengagement-reports/`.

[2] Aiyou Chen, Ruixuan Rachel Zhou, Joseph J. Lee, Nicholas Chamandy, and Henning Hohnhold. Confidence intervals for rate estimation with importance sampling in autonomous vehicle evaluation, 2026. arXiv:2604.03827.

[3] William G. Cochran. *Sampling Techniques*. John Wiley & Sons, 3rd edition, 1977.

[4] Christopher A. T. Ferro and Johan Segers. Inference for clusters of extreme values. *Journal of the Royal Statistical Society: Series B*, 65(2):545–556, 2003.

[5] Nidhi Kalra and Susan M. Paddock. Driving to safety: How many miles of driving would it take to demonstrate autonomous vehicle reliability? *Transportation Research Part A: Policy and Practice*, 94:182–193, 2016.

[6] Leslie Kish. *Survey Sampling*. John Wiley & Sons, 1965.

[7] Kung-Yee Liang and Scott L. Zeger. Longitudinal data analysis using generalized linear models. *Biometrika*, 73(1):13–22, 1986.

[8] Henry X. Liu and Shuo Feng. Curse of rarity for autonomous vehicles. *Nature Communications*, 15, 2024. Article 4808.

[9] Marcos López de Prado. *Advances in Financial Machine Learning*. John Wiley & Sons, 2018.

[10] National Highway Traffic Safety Administration. Standing general order 2021-01: Incident reporting for automated driving systems and level 2 advanced driver assistance systems, 2021. `https://www.nhtsa.gov/laws-regulations/standing-general-order-crash-reporting`.

[11] Jerzy Neyman and Elizabeth L. Scott. Statistical approach to problems of cosmology. *Journal of the Royal Statistical Society: Series B*, 20(1):1–43, 1958.

[12] Praprut Songchitruksa and Andrew P. Tarko. The extreme value theory approach to safety estimation. *Accident Analysis & Prevention*, 38(4):811–822, 2006.

[13] Maria A. Terres, Aiyou Chen, Ruixuan Rachel Zhou, and Claire M. McLeod. Behavioral event detection and rate estimation for autonomous vehicle evaluation. *Applied Stochastic Models in Business and Industry*, 2023. DOI: 10.1002/asmb.2769; also arXiv:2305.09936.

[14] Xingyu Zhao, Robab Aghazadeh-Chakherlou, Chih-Hong Cheng, Peter Popov, and Lorenzo Strigini. On the need for a statistical foundation in scenario-based testing of autonomous vehicles, 2025. arXiv:2505.02274.

[15] Lai Zheng and Tarek Sayed. Bayesian hierarchical modeling of traffic conflict extremes for crash estimation: A non-stationary peak over threshold approach. *Analytic Methods in Accident Research*, 2019.